

Predicting treated-water quality

Iteration 2 APPLIES RUN 2'S NEXT INVESTIGATION

GUARDRAILS · CHECKED BY THE RUN

- ✓ Leakage: no feature measured with the outcome
- ✓ Split: every test day later than every training day
- ✓ Baseline computed first (8.39); MAE 3.68 reported against it
- ✓ Single documented change vs run 2 (one feature)
- ✓ Target MAE ≤ 4.5 : **reached at 3.68** — ticked by the run

NEXT INVESTIGATIONS · THE SUGGESTIONS

- Try one **lag feature** (yesterday's output BOD) — the next domain candidate.
- Inspect the worst residual days against plant logs before trusting any alarm.

Iterations complete (3 of 3). The sign-off below is human.

Iterated from run 2: added **load** = **BOD × solids**. MAE improved 4.04 → 3.68 with the same guardrails — domain knowledge, not model size.

1 Question

Predict tomorrow's **output BOD (DBO-S, mg/L)** for the treatment plant, from today's inflow measurements and settings. Success metric: **MAE**. The baseline to beat: predict the historical mean.

2 Data

527 daily records, 12 variables, 92 missing cells (imputed with training-set medians). Schema follows the UCI Water Treatment Plant dataset (Poch 1993, CC BY 4.0); values regenerated offline for this session.

3 Features

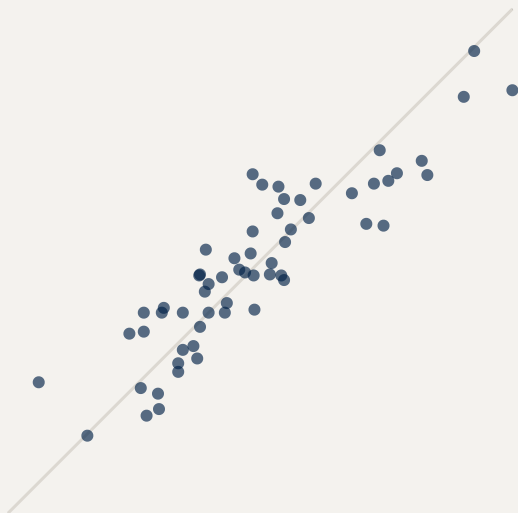
Q-E, PH-E, DBO-E, DQO-E, SS-E, SED-E, COND-E + load (engineered)

4 Model & split

Linear regression. Split: **time-aware + engineered load feature**.

5 Evaluation

MAE on test	3.68 mg/L
R ² on test	0.77
Baseline MAE (predict the mean)	8.39 mg/L



predicted vs actual, test set (perfect = diagonal)

✓ Human validation — to be ticked by the signatory

- ☐ The question is stated with a target and a unit
- ☐ Every feature exists at prediction time (no leakage)
- ☐ The split respects time: test days are strictly later
- ☐ The metric is reported on the test set only, next to the baseline
- ☐ Worst errors inspected; limits and uncertainty stated
- ☐ I approve this report for use within its stated limits

All runs precomputed for the session (no live computation on stage). Skill: skills/ml-analysis – swap in the real water-treatment.data with one line. Sources: UCI id 106; Sculley et al., NeurIPS 2015.

GUARDRAILS – THE NUMBERED LIST FROM PROMPT.TXT, VERIFIED BY THE RUN

G1	PASS	Target variable and unit stated — DBO-S, mg/L
G2	PASS	Imputation applied, training-set medians; imputed columns listed above
G3	PASS	No feature measured with or after the outcome — DQO-S excluded
G4	PASS	Every test day later than every training day; no shuffle
G5	PASS	Baseline computed first (8.39) and printed beside every score
G6	PASS	Limits and uncertainty stated
G7	PASS	Target reached — $\text{MAE } 3.68 \leq 4.5$, and G1–G6 all pass
H1	HUMAN	Human sign-off — left empty by the run

Scientist's panel — read before signing

ITERATION 2 · APPLIES RUN 2'S NEXT INVESTIGATION

Iterated from run 2: + load = BOD × solids.

GUARDRAILS CHECKLIST

- ✓ No feature measured with the outcome (DQO-S excluded)
- ✓ Time-aware split: every test day later than every training day
- ✓ Baseline computed first; every score reported next to it (8.39)
- ✓ Change vs run 2 is a single documented feature (load)
- ✓ Target $MAE \leq 4.5$ — reached at 3.68 · ticked by the run

LIMITS & UNCERTAINTY

Same validity bounds as run 2. The gain ($MAE\ 4.04 \rightarrow 3.68$) comes from one domain feature, not model complexity.

NEXT INVESTIGATIONS

- Try one lag feature (yesterday's output BOD) — next candidate from the domain.
- Check the worst residual days against plant logs before trusting alarms.

Sign-off (human only): approved by _____ date _____